

ai

Josh Oltmanns

ai (or stuff related to it that I  
have heard of and use  
sometimes)

Josh Oltmanns

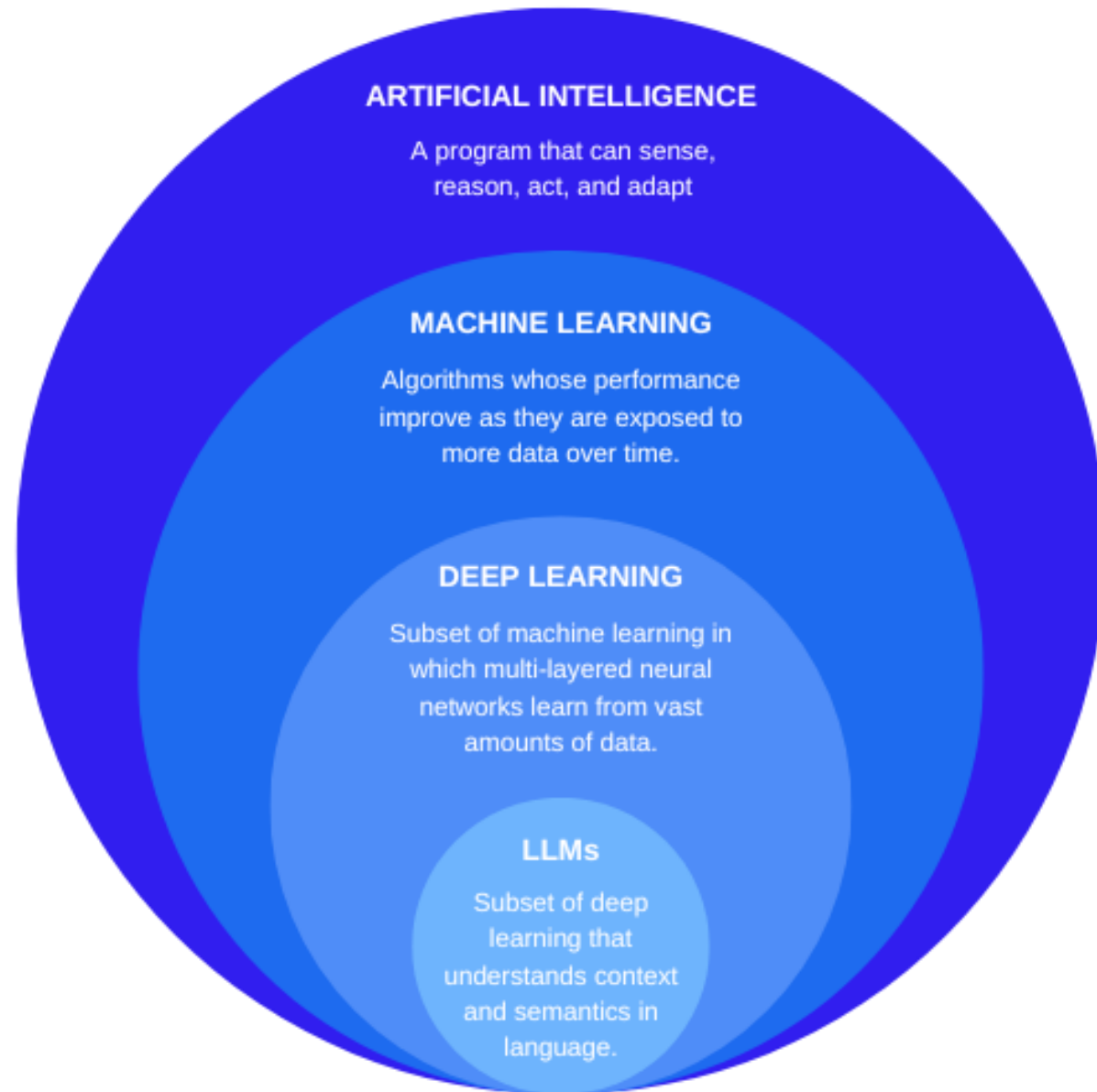
# paper and code examples

- our intro to LLMs paper:
- Brickman, J., Gupta, M., & Oltmanns, J. R. (2025). Large language models for psychological assessment: A comprehensive overview. *Advances in Methods and Practices in Psychological Science*.  
[https://doi.org/10.31234/osf.io/qm9ae\\_v1](https://doi.org/10.31234/osf.io/qm9ae_v1)
- accompanying coding examples:
- <https://github.com/mehak25/Intro-to-LLM>

# why am i interested in ai?

- assessment
  - sound measurement is at the heart of psychological science.
  - most of what we know relies on self-report assessment.
    - self-report has obvious weaknesses/blind spots.
  - multimethod assessment is #goals for psychologists (APA, 2020).
    - it is rarely incorporated into studies and treatment.
    - ai provides opportunities to incorporate multimethod assessment.
- lots more reasons
  - improve ability to interpret complex data/images.
  - provide scalable evidence-based treatment.
  - improve ability to predict outcomes.
  - provide nuanced insights into psychological constructs.
  - ensure safe use and reduce bias.

# ai, ML, LLM



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i can tell you a  
little about this  
one



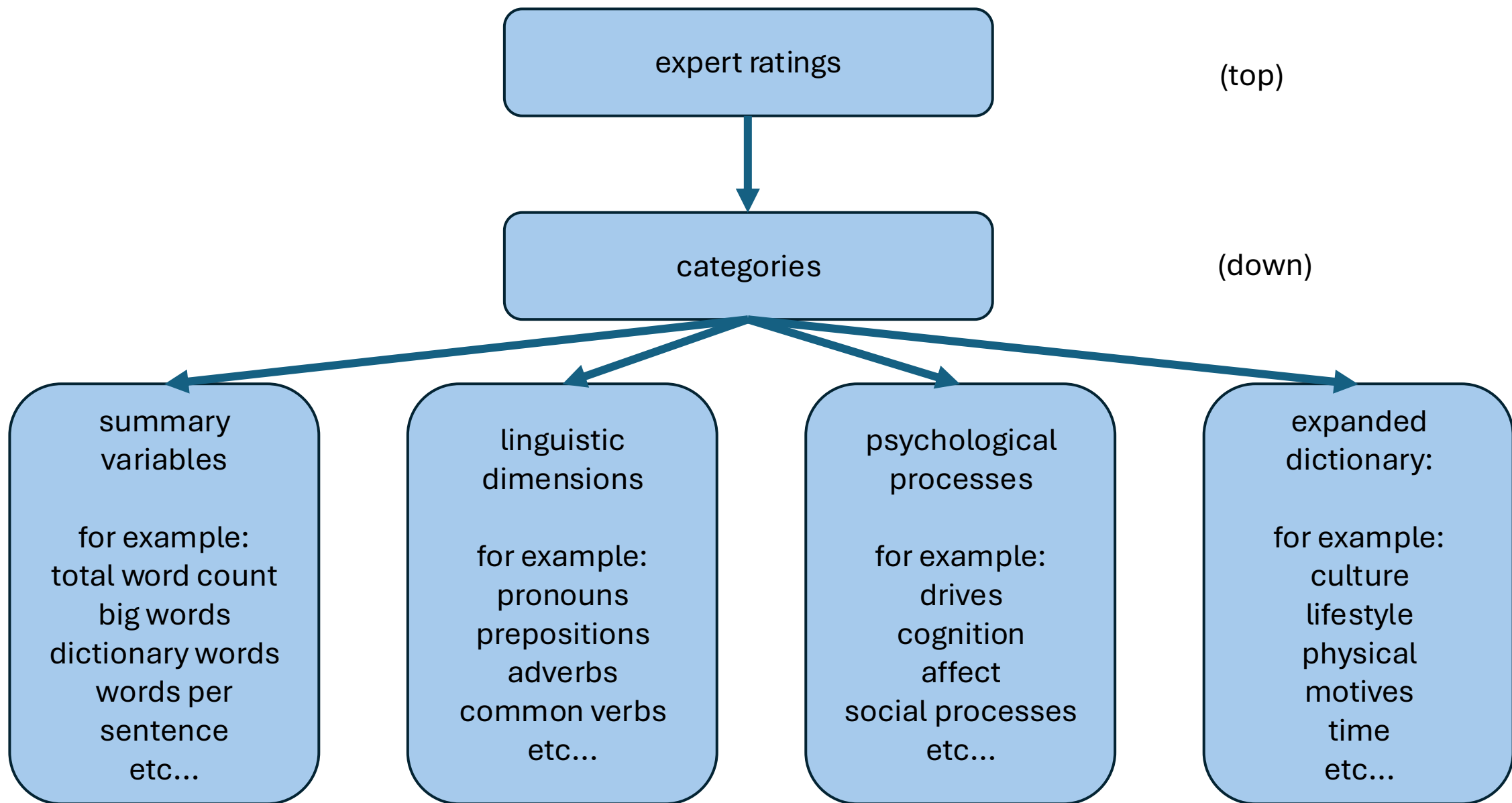
# natural language processing

- language has several potential strengths as an assessment tool :
  - behavioral
  - natural (sometimes)
  - rich
  - scalable
  - can enhance assessment coverage
  - is literally where the Big 5 came from

# natural language processing

- linguistic inquiry and word count (LIWC)
  - developed by Dr Jamie Pennebaker (UT-Austin).
    - President-elect, Association for Psychological Science (APS)
  - human raters (e.g., psychologists) organized ratings of words in texts into different categories.
  - these categories can then be scored from texts, providing psychological or linguistic insights from text.
    - this is called a “bag of words” approach.



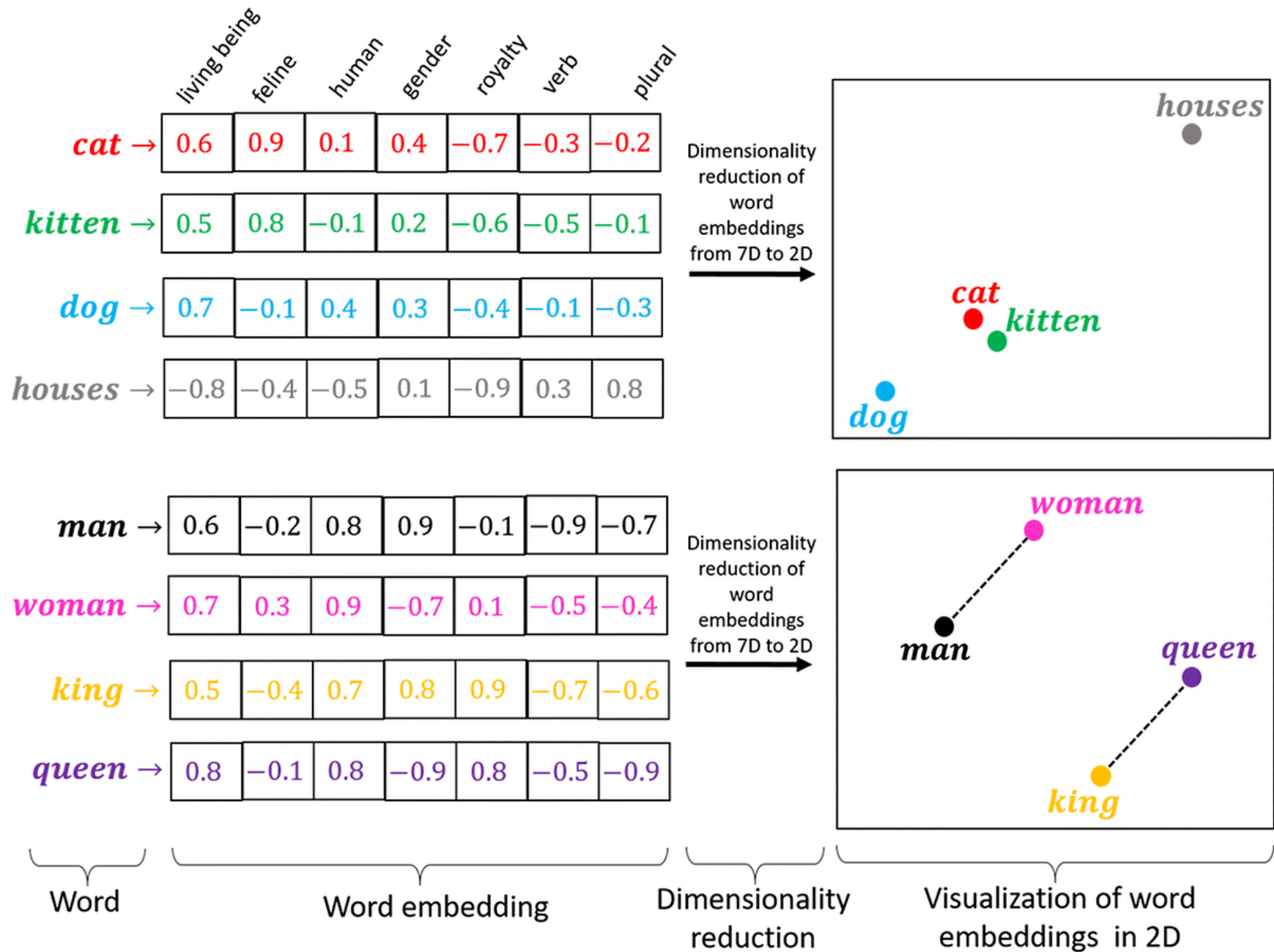


# LIWC limitations

- scores are percentages of words in a text in the categories.
  - “Bag of words” approach.
- human judgment (top down approach) not comprehensive and may miss word meanings.
- may miss words or misrepresent their importance.
- no context - may miss figures of speech, irony, negations.
  - no real sense of meaning.

# the word embedding

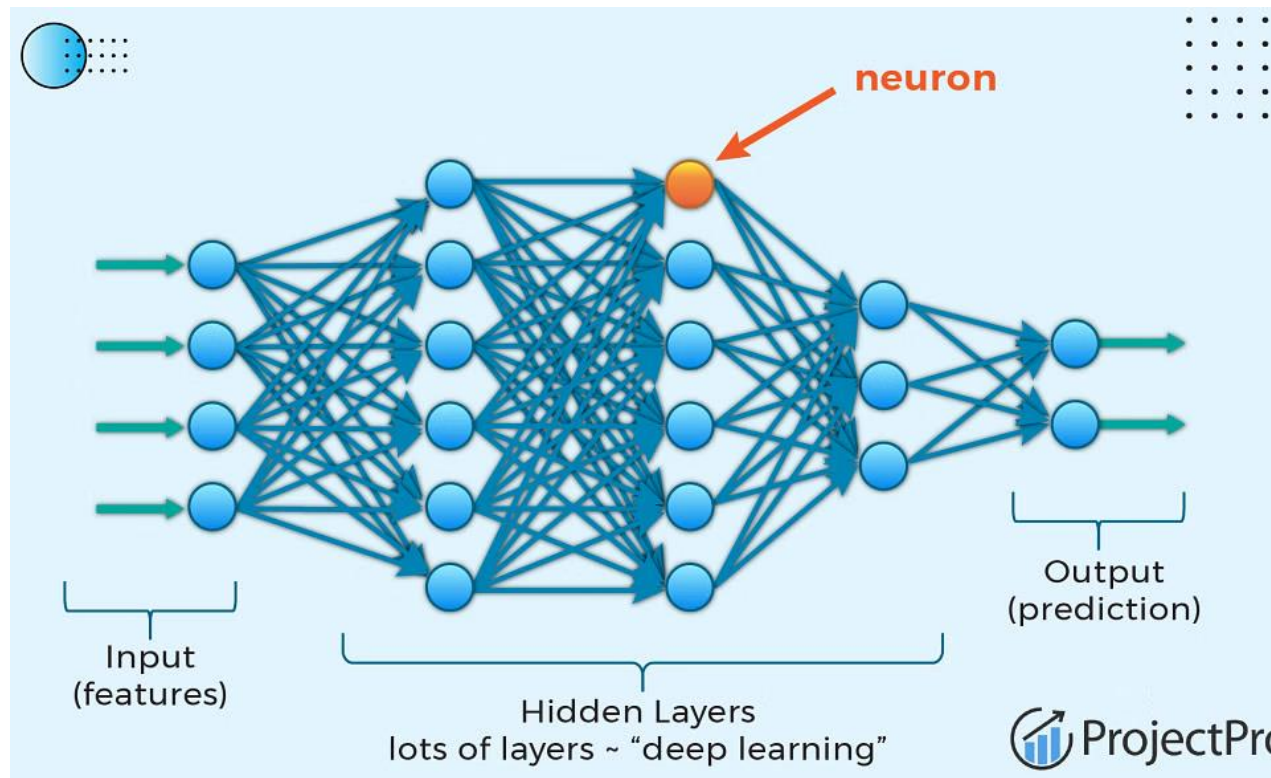
- word2vec
  - words in documents converted into vectors (Mikolov et al., 2013)
  - used machine learning to develop word embeddings.
    - training text “corpus”: google news dataset (~100 billion words), Wikipedia, other web data.
    - skip gram (predict surrounding words from current word)
    - continuous bag of words (predict current word from surrounding words)



# the word embedding

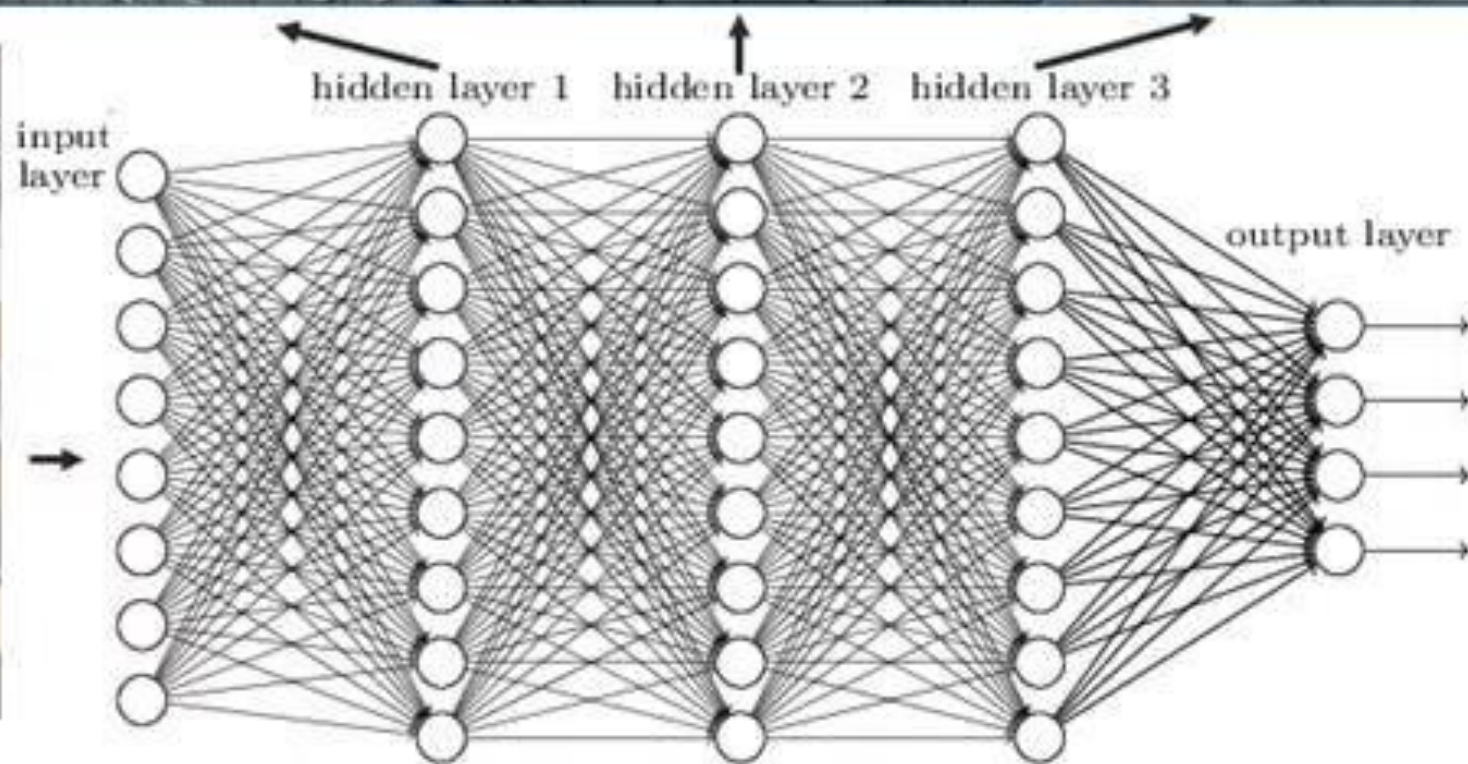
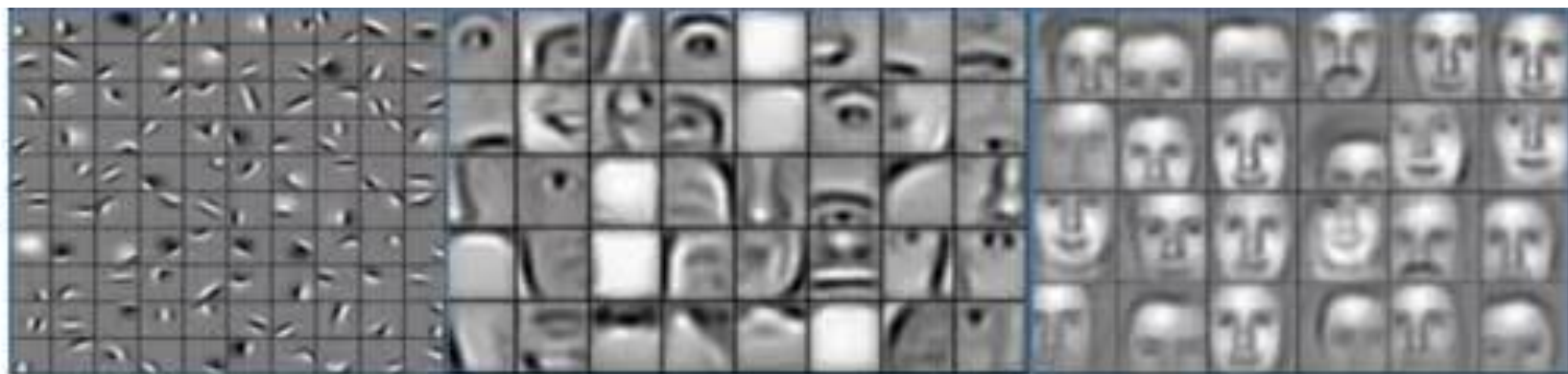
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    - continuous bag of words (i.e., predict current word from surrounding words)
  - quantitative, multidimensional representation of *meaning*
  - static.

# deep learning



- Advanced type of machine learning inspired by how humans process information. Learns patterns by processing data through multiple layers, allowing for increasingly more complex connections.

Deep neural networks learn hierarchical feature representations

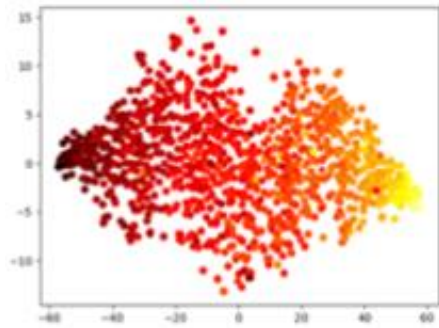


# the word embedding

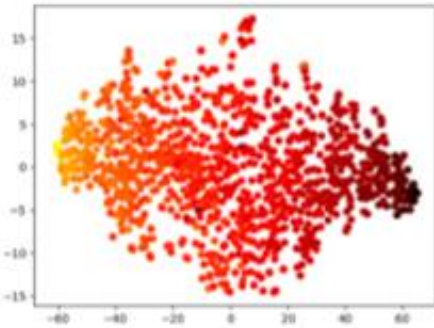
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  - quantitative, multidimensional representation of *meaning*
  - static.
- ELmo
  - “Embeddings from Language Models” (Peters et al., 2018)
  - deep learning (recurrent neural networks and long short term memory).
    - predict words sequentially moving forwards and backwards.
  - contextualized embeddings
    - embeddings change depending on word context.

# the word embedding

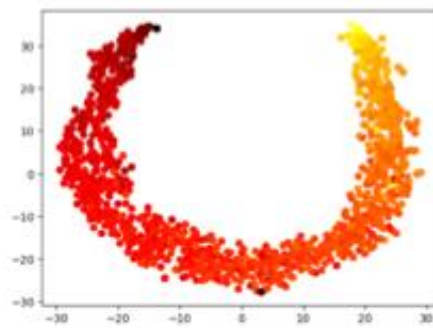
- you can also get sentence and document-level embeddings.
- this can be helpful for visualizing semantic meaning of texts, for example:



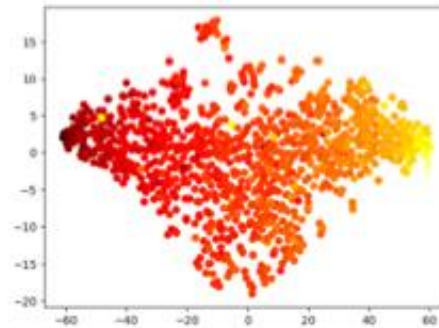
Neuroticism



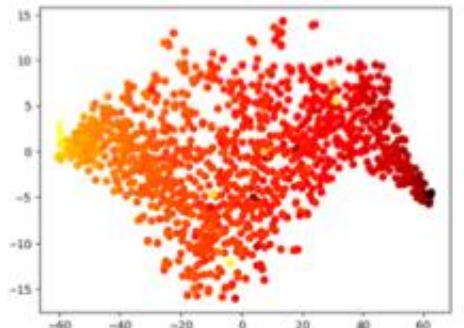
Extraversion



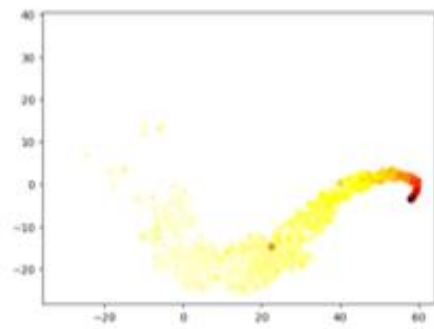
Openness



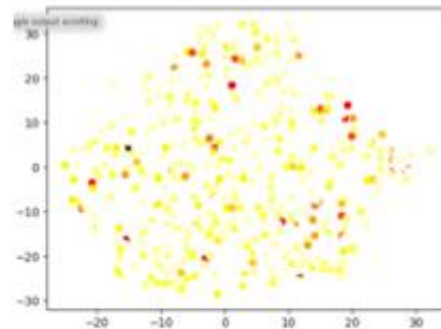
Agreeableness



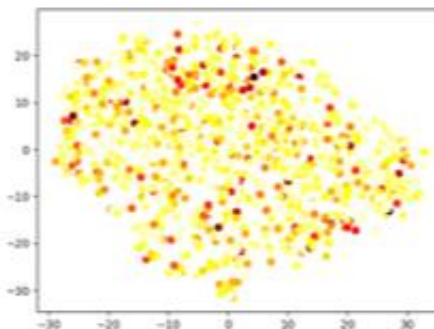
Conscientiousness



Borderline



Schizoid



Obsessive-Compulsive

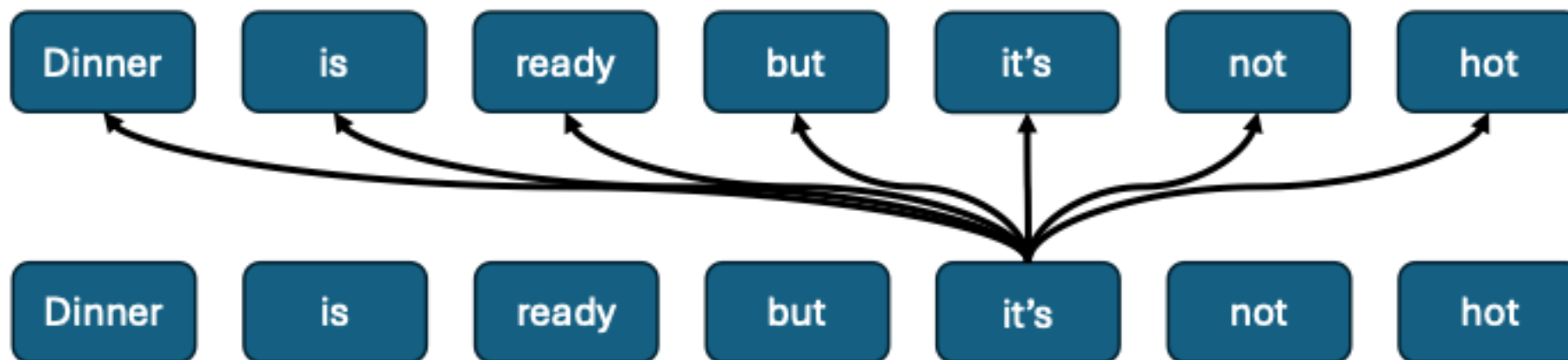
# transformer model

- RNNs and LSTMs process sequences step by step, making it hard to capture long-range dependencies and difficult to parallelize.
- transformers introduced *self-attention* to model all positions in a sequence simultaneously, enabling better context modeling and faster training.
- key advance: self attention.

# transformer model



a. Sequential attention



b. Self-attention

# **transformer model**

- self-attention mechanism
- faster – simultaneous attention to other words in sentence.

# large language models (LLMs)

- transformer allowed for faster processing of massive data.
  - model sizes could grow (and be better at the same time).
- Bidirectional Encoder Representations from Transformers (BERT)
  - Devlin et al. (2017)
  - Trained on BooksCorpus (11k books) and English Wikipedia.
    - using training techniques: word masking and next sentence prediction.
  - Final model – millions of parameters.
    - Although not technically big enough to be an “LLM” – so just “LM”
  - Lots of updated versions, multilingual, fine-tuned for specific purposes.
  - RoBERTa (Liu et al., 2019).
    - Updated with more training data and optimized training tasks.
- “pretrained” – training on massive amounts of data, usually web-based.

# large language models (LLMs)

| Model Family    | Parameters          | Developer                       | API                            | Open Source | License           | Training Data                                                                                                        | Max Token Limit |
|-----------------|---------------------|---------------------------------|--------------------------------|-------------|-------------------|----------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>BERT</b>     | 4M - 340M           | Google (Devlin et al., 2019)    | Hugging Face                   | Yes         | Apache 2.0        | BookCorpus and English Wikipedia                                                                                     | 512             |
| <b>GPT</b>      | Estimated 8B - 1T+  | OpenAI (Achiam et al., 2023)    | OpenAI API                     | No          | Proprietary       | Publicly available and licensed data                                                                                 | 128k            |
| <b>Llama</b>    | 1B - 405B           | Meta                            | Llama API, Hugging Face        | Yes         | Custom            | Publicly available online data                                                                                       | 128K            |
| <b>Claude</b>   | Estimated 70B - 2T+ | Anthropic                       | Anthropic API                  | No          | Proprietary       | Public internet information, non-public third-party data, internally created data; fine-tuned with Constitutional AI | 1M              |
| <b>Gemini</b>   | 1.8B - 1T+          | Google DeepMind                 | Gemini API, Google Vertex AI   | No          | Proprietary       | Publicly available filtered web data, licensed 3 <sup>rd</sup> party data                                            | 2M              |
| <b>Falcon</b>   | 1B - 180B           | Technology Innovation Institute | Hugging Face                   | Yes         | Apache 2.0        | RefinedWeb, filter of Common Crawl dataset                                                                           | 32K             |
| <b>Mistral</b>  | 7B - 120B+          | Mistral AI (Jiang et al., 2023) | Mistral AI, Hugging Face       | Some        | Apache 2.0        | Open web data                                                                                                        | 128K            |
| <b>DeepSeek</b> | 7B - 671B           | Hangzhou DeepSeek AI            | DeepSeek API, Hugging Face     | Some        | Apache 2.0 (most) | Web data, code, math                                                                                                 | 128K            |
| <b>Qwen</b>     | 0.5B - 110B         | Alibaba Cloud                   | Alibaba Cloud, Hugging Face    | Some        | Apache 2.0        | Public web documents, encyclopedia, books, math, code, synthetic data                                                | 1M              |
| <b>Gemma</b>    | 1B - 27B            | Google DeepMind                 | Google Vertex AI, Hugging Face | Yes         | Apache 2.0        | Web documents, code, and math                                                                                        | 128K            |

# large language models (LLMs)

- useful tasks:
  - summarization
  - text classification
  - named entity recognition
  - question answering
  - sentiment analysis
  - sentence completion
  - language translation

# some techniques for using LLMs

1. feature extraction
2. fine-tuning
3. prompt engineering

# feature extraction

- obtain contextualized embeddings from your text using an LLM.
  - use these embeddings as predictor variables in downstream analysis (e.g., regression).
- Wulff & Mata (2023)
  - examine contextualized embeddings from items across personality questionnaires to examine the jingle/jangle fallacy.
  - e.g., items in different scales across measures may have same semantic meaning; items in same scales across measures may have different semantic meaning
- Abdurahman et al (2024)
  - use contextualized embeddings and individual scores on self-report personality items to predict scores on previously unseen items.

# fine-tuning

- further training pretrained LLMs with more specific data.
- model parameters are updated to reflect domain specific language
  - e.g., language related to personality or mental disorders.
- can be effective with smaller datasets.
- computationally expensive.
- uses traditional machine learning techniques (e.g., cross validation, hyperparameter tuning).
- Gul et al. (2024) fine-tuned 3 LLMs to predict political beliefs from social media language.
- Ohse et al. (2023) fine-tuned BERT and GPT 3.5 to predict depression from depression interview language.

# prompt engineering

- designing prompts to guide the output of LLMs, enabling improved performance without retraining the model.
- less computationally expensive.
- “hard prompts”
  - zero-shot, one-shot, few-shot
- “soft prompts”
  - set of trainable embeddings added to the input text.
  - soft prompt trained with labeled examples – supervised learning.
  - less computationally expensive than fine-tuning and can work better if using very large models (Peng et al., 2024).

## MADRS Assessment Prompt Structure

### Task Description:

Analyze a diarized transcript of a psychiatric session where the Montgomery-Åsberg Depression Rating Scale (MADRS) questionnaire is being administered. Predict the rating (0-6) that the practitioner would likely give for the specified MADRS item based on the patient's responses and the conversation.

### MADRS Item Components:

- **Item Name:** Reported Sadness
- **Description:** Represents reports of depressed mood, regardless of whether it is reflected in appearance or not. Includes low spirits, despondency or the feeling of being beyond help and without hope.
- **Key Questions:**
  - In the past week, have you been feeling sad or unhappy?
  - Does the feeling lift at all if something good happens?
  - How much of each day? How many days this week?

### Rating Scale (0-6):

- **0:** Occasional sadness in keeping with circumstances
- **2:** Sad or low but brightens up without difficulty
- **4:** Pervasive feelings of sadness or gloominess
- **6:** Continuous or unvarying sadness, misery
  - (Odd numbers represent intermediate states)

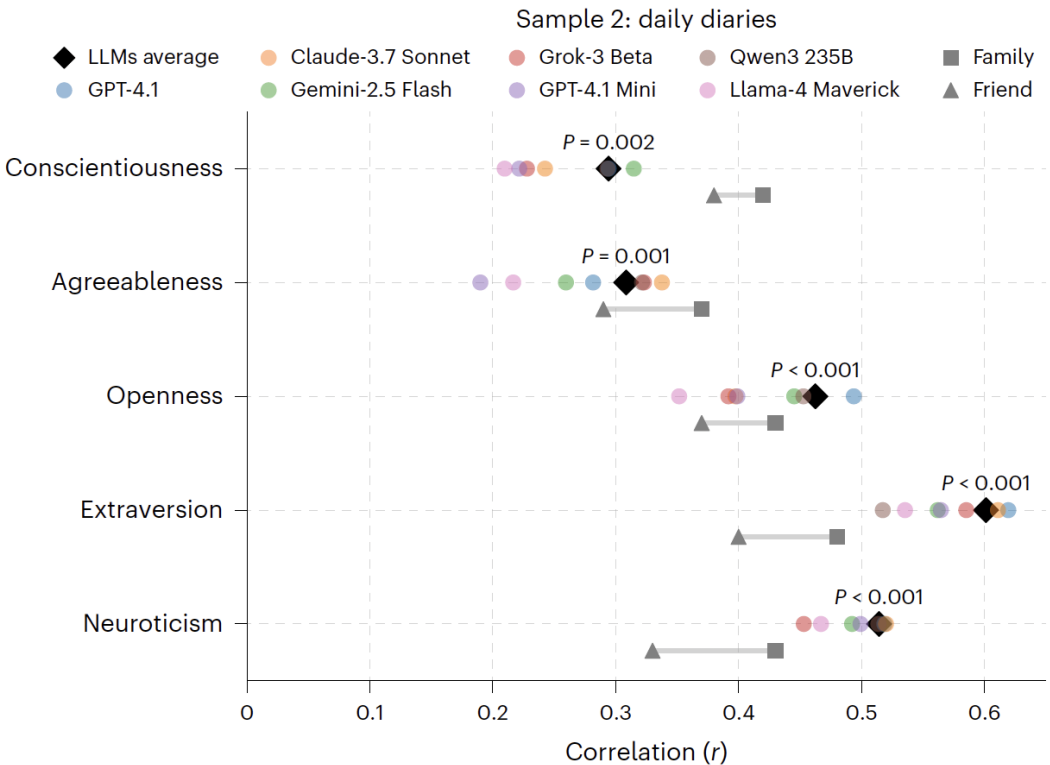
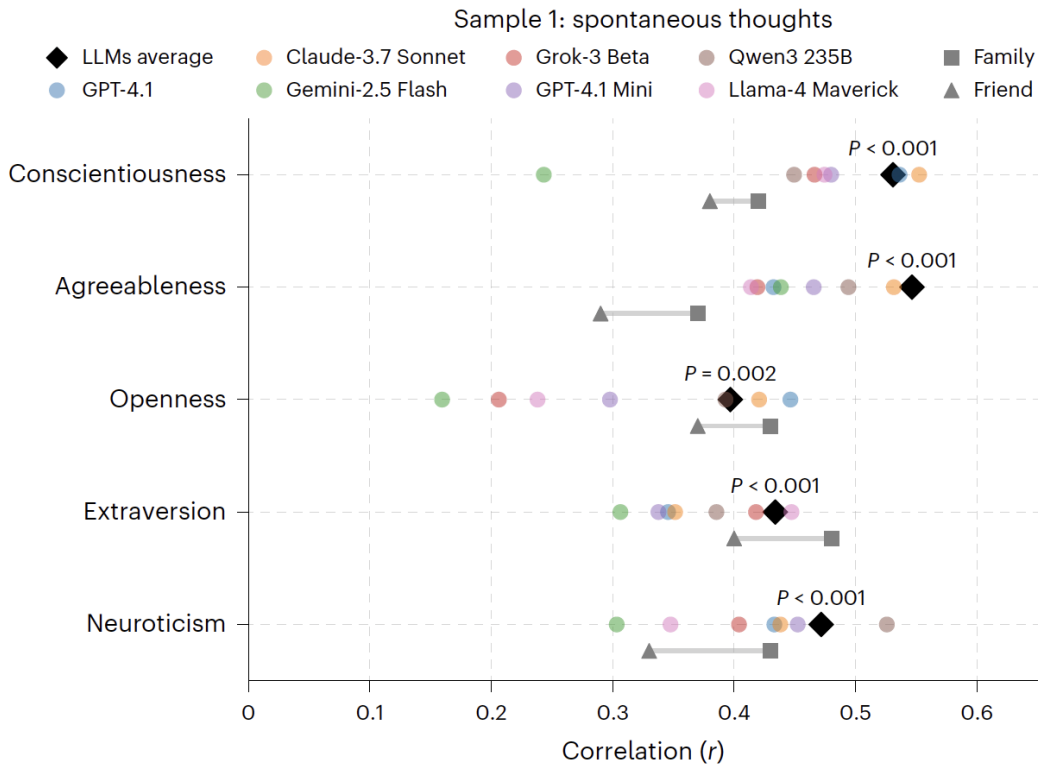
### Required Output Format:

Rating: [0-6]  
Explanation: [2-3 sentences]  
Key Utterances: [relevant lines]  
Most Relevant Question: [from transcript]

Table 3: Performance Comparison of Large Language Models for MADRS Total Score Prediction

| Cont. Len.            |      | MAE          |             | R <sup>2</sup> |             |
|-----------------------|------|--------------|-------------|----------------|-------------|
|                       |      | Full         | Segmented   | Full           | Segmented   |
| Qwen 2.5 Inst. (72B)  | 128K | 4.90 ± 0.11  | 3.69 ± 0.05 | 0.69 ± 0.03    | 0.84 ± 0.00 |
| Llama 3.1 Inst. (70B) | 128K | 6.12 ± 0.17  | 4.86 ± 0.18 | 0.54 ± 0.05    | 0.74 ± 0.03 |
| Qwen 2 Inst. (72B)    | 128K | 7.10 ± 0.82  | 4.40 ± 0.20 | 0.40 ± 0.16    | 0.78 ± 0.03 |
| Qwen 2.5 Inst. (32B)  | 128K | 15.55 ± 0.20 | 3.52 ± 0.17 | −3.74 ± 0.08   | 0.85 ± 0.02 |
| Qwen 2.5 Inst. (14B)  | 128K | 15.61 ± 0.19 | 3.62 ± 0.17 | −3.80 ± 0.05   | 0.84 ± 0.03 |
| Qwen 2.5 Inst. (7B)   | 128K | 17.36 ± 0.22 | 4.47 ± 0.19 | −4.05 ± 0.04   | 0.77 ± 0.03 |
| Qwen 2.5 Inst. (3B)   | 32K  | 19.40 ± 0.24 | 7.03 ± 0.21 | −4.45 ± 0.06   | 0.50 ± 0.04 |
| Llama 3.1 Inst. (8B)  | 128K | 19.42 ± 0.17 | 9.96 ± 0.22 | −2.27 ± 0.08   | 0.06 ± 0.04 |

**Note:** MADRS = Montgomery-Åsberg Depression Rating Scale; MAE = Mean Absolute Error. Model parameters (B) are shown in billions. Best performing metrics are highlighted in **bold**. Negative R<sup>2</sup> values indicate poor model fit relative to baseline.



**Fig. 2 | Correlations between LLM estimates of personality from stream of thought prompts (Study 1, left panel) and daily diaries (Study 2, right panel) and self-report survey compared with meta-analytic estimates of self–other survey agreement (Connelley and Ones, 2010).** For all but conscientiousness assessment based on daily diaries, LLM-based estimates match or exceed typical self–other agreement benchmarks for family and friends (grey boxes and

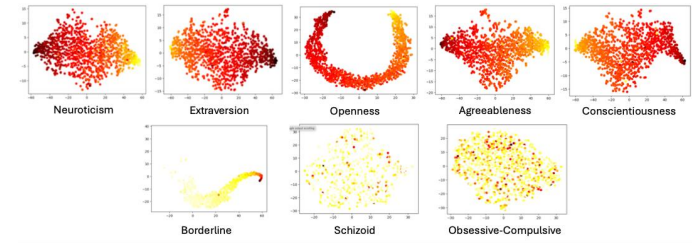
triangles, respectively). Values in the figure represent Pearson correlations ( $r$ ). In Sample 1 ( $N = 60$ ,  $df = 58$ ), values of  $r \geq 0.42$ ,  $r \geq 0.33$  and  $r \geq 0.25$  are significant at  $P < 0.001$ ,  $P < 0.01$  and  $P < 0.05$ , respectively, using two-sided tests. In Sample 2 ( $N = 108$ ,  $df = 106$ ), all values of  $r \geq 0.31$ ,  $r \geq 0.25$  and  $r \geq 0.19$  are significant at  $P < 0.001$ ,  $P < 0.01$  and  $P < 0.05$ , respectively, using two-sided tests. No adjustments were made for multiple comparisons.

# choosing LLMs

- training data – what was it trained on?
- text limits – how much text can it accommodate at one time?
- size – how many parameters?
- usage limits – is it free?
- model ‘cards’ – one place, all models details on these things.

# visualization

- black box problem in deep learning.
- t-SNE to visualize embeddings ->
- visualizing attention weights (Vig, 2019).



|             |  |                                                                    |
|-------------|--|--------------------------------------------------------------------|
| High<br>OPE |  | trippy day ahead ....                                              |
|             |  | nothingness at last ....                                           |
|             |  | shutter island was good .. .                                       |
| High<br>CON |  | they are over ... yah                                              |
|             |  | my phone is not working ...                                        |
|             |  | stoked on the exam schedule !                                      |
| High<br>EXT |  | 40 % math midterm ? thank god 3/4 count .                          |
|             |  | got a co-op job interview ! woo !                                  |
|             |  | just had some damn good pears . note to self : buy more ? damnit . |
| Low<br>AGR  |  | found free bag of skittles in the vending machine , jackpot .      |
|             |  | at the beach with keira ! ! !                                      |
|             |  | getting ready for brittany's dance recital tonight ! !             |
| High<br>NEU |  | had fun at nathans barmitzvah last night ! ! !                     |
|             |  | i have made 72 cupcakes in the last 3 days ! ! ! ! lol             |
|             |  | just finished my science project :) )                              |
| Low<br>AGR  |  | sooo excited for new school year :) going to make it awesome       |
|             |  | grudges are so ridiculous and pointless ı - ı                      |
|             |  | ahh shit almost 1 ! ? i need to finish this paper ! ! !            |
| High<br>NEU |  | that sure was a fun ride home O.o                                  |
|             |  | wants to just skip to the next weekend .                           |
|             |  | can't believe i got that done in time .....                        |
| High<br>NEU |  | packing to go back to school makes me sad .                        |
|             |  | losing things and is getting extremely frustrated . :(             |
|             |  | is amazed at how similar cameras are to your eyes .                |
| High<br>NEU |  | whhhaaa ? it's only wednesday ...                                  |
|             |  |                                                                    |
|             |  |                                                                    |

# visualization

- black box problem in deep learning.
- t-SNE to visualize embeddings (like the figure I showed earlier).
- visualizing attention weights (Vig, 2019).
- attention heatmap.

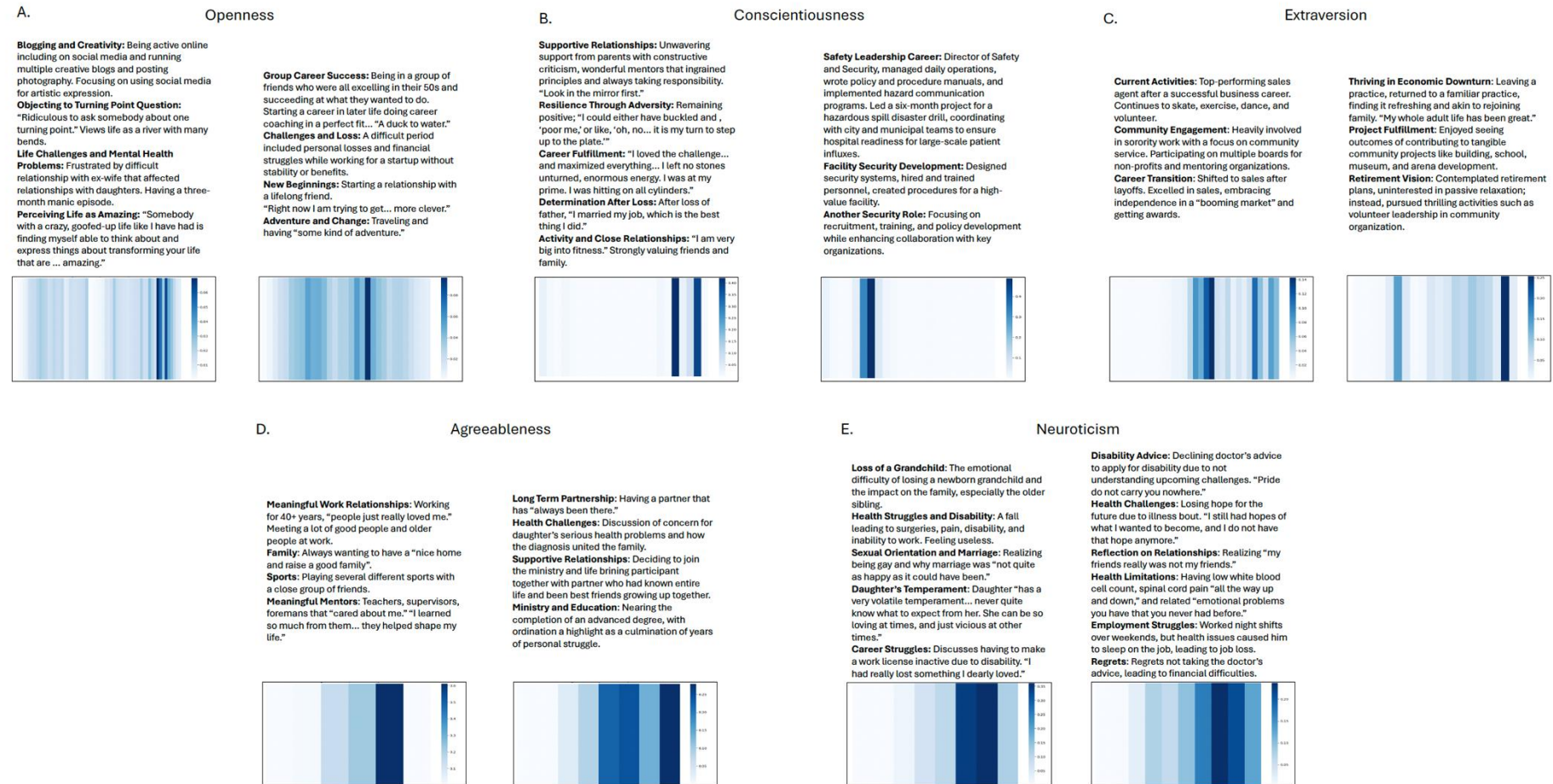


Figure 3: Heatmap showing attention given to different text windows in the whole transcript for each personality trait.

# ethical considerations

- “stochastic parrots” (Bender et al., 2021).
- bias across age, race, gender, other demographics (Raza et al., 2024).
- techniques may reduce model biases including data augmentation, bias correction algorithms, and fairness metrics

# researchers using LLMs must be...

- 1) aware of bias, especially that is directly relevant to their area of research
- 2) active in ensuring fairness in model development (e.g., comparing model results and predictions across various groups)
- 3) transparent about the existence of biases in the models that they use (e.g., describing the biases and their potential influence on the results in Discussion sections)
- 4) constantly updating their LLM use in accordance with the latest techniques to reduce harm
- 5) supporting or collaborating with researchers from minority groups, especially those that might be a focus of the research.

# construct validation techniques

- for example, fine-tuned model prediction should be validated as in prior research with convergent, discriminant, and criterion validity tests, etc. (Strauss & Smith, 2009).
- examine the nomological network of model predictions.
- internal consistency, test-retest reliability.

# interdisciplinary collaboration

- collaboration is essential, e.g., with computer scientists.
- mutual respect, requiring time and effort.

# conclusion

- ai has the power to revolutionize psychological research.
- relatively little is known to-date.
- studies are limited in sample size, diversity, and measurement.
- awareness of bias, strong psychometric validation techniques, and interdisciplinary collaboration will be essential for advancement.

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